

Throughput-Delay Trade-off and Systems Research in Sensor Networks

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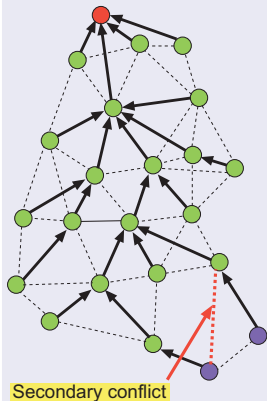
Outline

- 1 Maximizing Convergecast Throughput [MASS '09, TMC '10 (rev)]
 - Motivation and Preliminaries
 - Scheduling with Unlimited Frequencies
 - Approximations on Random Geometric Graphs
- 2 Throughput-Delay Trade-off [Infocom '10 (wrksh), TON '10 (sub)]
 - Motivation
 - Bi-criteria Formulation
 - Optimal Spanning Trees
- 3 Industrial / Systems Research [ACM MC2R '10]
 - Siemens Corporate Research
 - IBM Research
 - Other Industry Experience

Outline

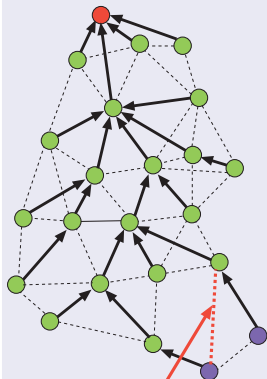
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Model and Assumptions



e.g., Applications with fast & efficient data collection requirements.

Model and Assumptions



Secondary conflict

e.g., Applications with fast & efficient data collection requirements.

- TDMA periodic scheduling
(- every node generates a single packet at the beginning of each frame
- every node transmits once per slot)
- Graph-based network & interference model
- Half-duplex, single transceiver
- Receiver-based Frequency Assignment
- Non-interfering orthogonal channels
- In-network aggregation
(distributive, algebraic
- aggregation happens at each hop)



Schedule length

Convergecast and Benefits of Multiple Frequencies

Convergecast

A **many-to-one** communication pattern (i.e., flow of data): from all the nodes to a sink (opposite of broadcast).

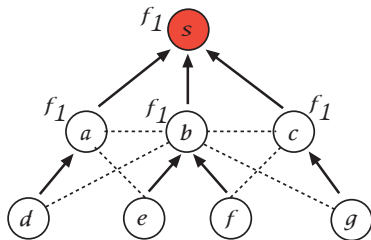


Figure: Single frequency.

Convergecast and Benefits of Multiple Frequencies

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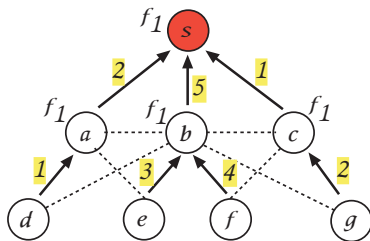


Figure: Schedule length is 5.

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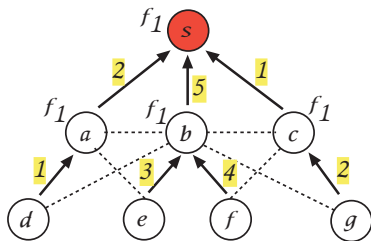


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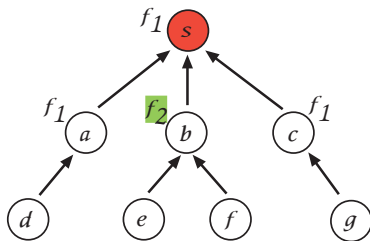


Figure: Two frequencies.

Convergecast and Benefits of Multiple Frequencies

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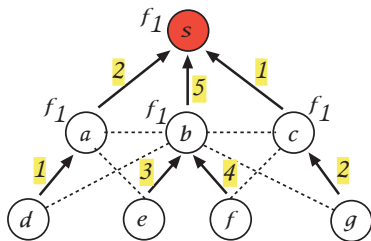


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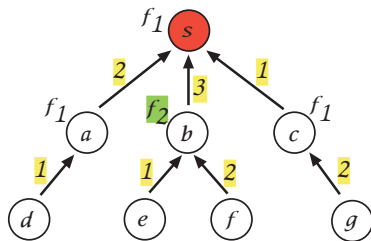


Figure: Schedule length is 3.

Convergecast and Benefits of Multiple Frequencies

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A **many-to-one** communication pattern (i.e., flow of data): from all the nodes to a sink (opposite of broadcast).

	Frame 1			Frame 2		
Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
<i>s</i>						
<i>a</i>						
<i>b</i>						
<i>c</i>						

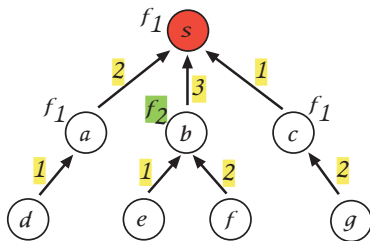


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Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
s	<i>c</i>	<i>a d</i>	<i>b e f</i>	<i>c g</i>	<i>a d</i>	<i>b e f</i>
a	<i>d</i>			<i>d</i>		
b	<i>e</i>	<i>f</i>		<i>e</i>	<i>f</i>	
c		<i>g</i>			<i>g</i>	

Pipelining

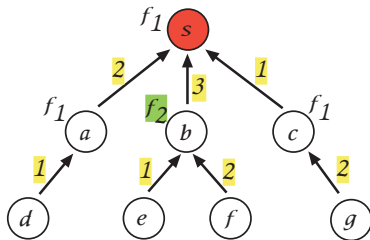


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Problem Formulation

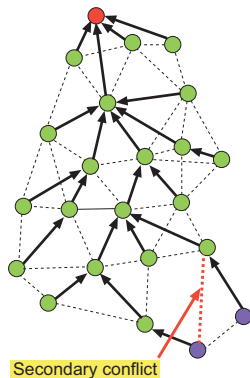
Joint Frequency and Time Slot Assignment

Given a network $G = (V, E)$ of arbitrarily deployed nodes, K orthogonal frequencies, and a routing tree $T \subseteq G$ rooted at the sink $s \in V$

- Assign a frequency to each receiver in T
- Assign a time slot to each edge in T

such that the aggregated throughput at the sink is maximized, where

Aggregated Throughput = $1/\text{Schedule Length}$
(equivalent to minimizing the Schedule Length)



Scheduling Complexity

Scheduling Complexity

Theorem

Joint Frequency and Time Slot Assignment is NP-hard on general graphs.

Proof: Reduction from Vertex Color.

Questions

- Does the problem still remain NP-hard with unlimited frequencies?
- If not, how many frequencies (at a min, or at most) are needed?
- Can we design algorithms that have provable guarantee?

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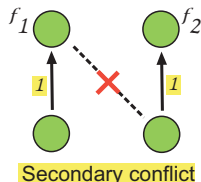
- Does the problem still remain NP-hard with **unlimited** frequencies?
- If not, how many frequencies (at a min, or at most) are needed?
- Can we design algorithms that have **provable** guarantee?

Theorem

Given a spanning tree T on a graph $G = (V, E)$, find the **minimum** number of frequencies required to remove **all** secondary conflicts?

Minimum Frequency Assignment is NP-hard.

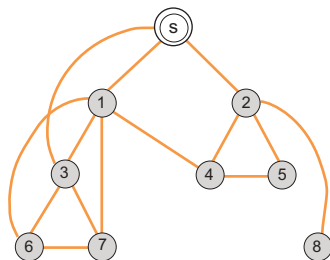
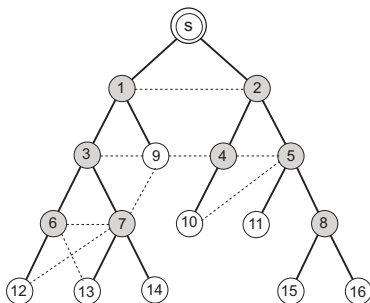
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A Frequency Upper Bound

Constraint Graph

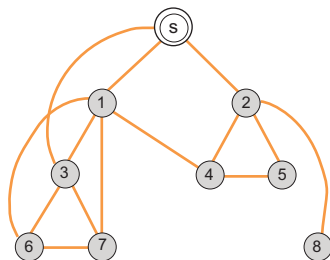
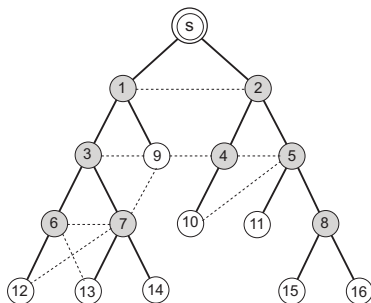
- For each receiver in G create a node in G_C .
- Create an edge between two nodes in G_C if the corresponding receivers in G form a secondary conflict.



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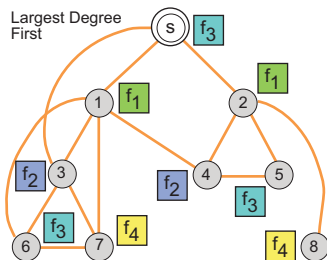
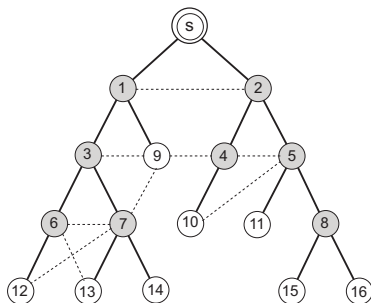
Lemma

$K_{max} \leq \Delta(G_C) + 1$, where $\Delta(G_C)$: max degree in G_C . (Vertex Color)

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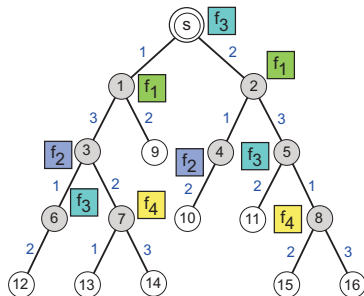
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Time Slot Assignment

Algorithm: BFS-TIME SLOT ASSIGNMENT

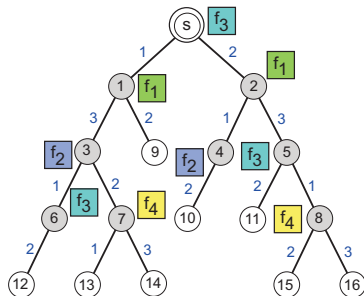
1. **while** $E_T \neq \phi$ **do**
2. $e \leftarrow$ next edge from E_T in BFS order
3. Assign minimum time slot to e respecting adjacency constraint
4. $E_T \leftarrow E_T \setminus \{e\}$
5. **end while**



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Theorem

BFS-TIME SLOT ASSIGNMENT gives a schedule of minimum length $\Delta(T)$, where $\Delta(T)$: maximum node degree in T .

Proof: By induction.

So far...

Joint Frequency and Time Slot Assignment is

- **NP-hard** with **limited** (constant number) of frequencies on arbitrary graphs.
- **Polynomial** with **unlimited** (at most $\Delta(G_C) + 1$) frequencies for arbitrary graphs.

So far...

Joint Frequency and Time Slot Assignment is

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Next...

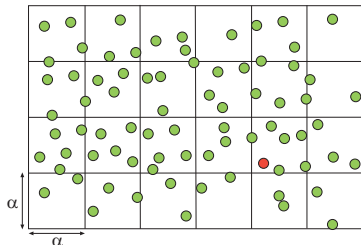
Approximation algorithms for **Random Geometric Graphs** with **limited** (constant number) frequencies.

- When nodes have uniform transmission range (**Unit Disk Graphs**)
- When nodes have non-uniform transmission range (**Disk Graphs**)

Approximation on UDG

Algorithm $\mathcal{A}_{UDG}$ Overview

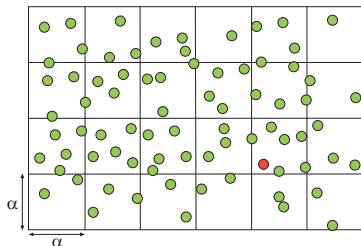
- **Phase 1:** Assign frequencies to the receivers in each cell such that the **maximum** number of nodes transmitting on the same frequency (**load**) is minimized.
(Load-Balanced Frequency Assignment)
- **Phase 2:** Assign time slot greedily to each edge.



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Theorem

Algorithm $\mathcal{A}_{UDG}$ gives a **constant factor** $8\mu_\alpha \cdot \left(\frac{4}{3} - \frac{1}{3K}\right)$ approximation on the optimal schedule length, where $\mu_\alpha > 0$ is a constant for any given cell size $\alpha \geq 2R$, and given node deployment, where K is the number of frequencies.

Phase 1: Frequency Assignment

Lemma

Load-Balanced Frequency Assignment is NP-hard.

Proof: Reduction from Job Scheduling on identical machines [Garey, Johnson '79].

Phase 1: Frequency Assignment

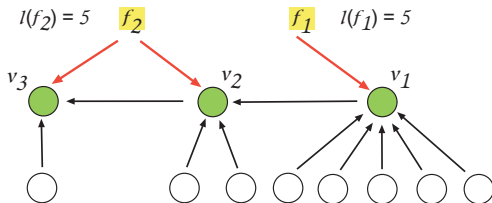
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Algorithm: FREQUENCYGREEDY

- 1 Sort the receivers in **non-increasing** order of **in-degrees**:
 $\deg^{in}(v_1) \geq \deg^{in}(v_2) \geq \dots \geq \deg^{in}(v_n)$
- 2 Starting from v_1 , assign each successive receiver a frequency that has the **least load**, breaking ties arbitrarily



Phase 1: Frequency Assignment

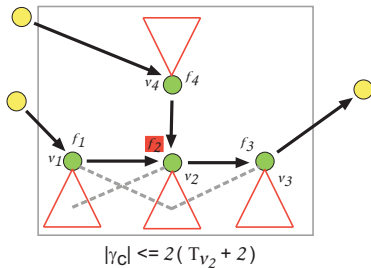
Lemma

FREQUENCYGREEDY gives a $\left(\frac{4}{3} - \frac{1}{3K}\right)$ approximation on the min-max load, where K is the number of frequencies.

Proof: Reduction from Graham's list scheduling according to **Longest Processing Time** first [Graham '69].

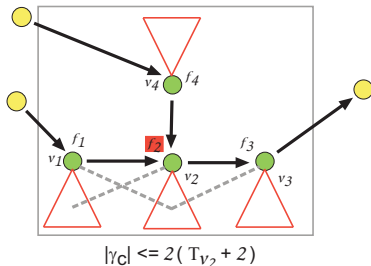
Phase 2: Time Slot Assignment

In each cell:



Phase 2: Time Slot Assignment

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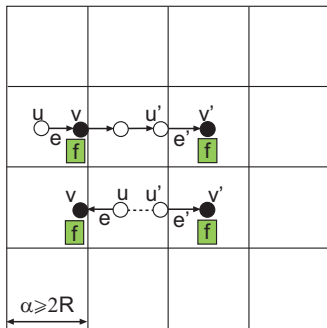
Lemma

Suppose L_c^{FG} denote the load on the **maximally loaded frequency** in cell c achieved by **FREQUENCYGREEDY**.

Then, all edges in c can be scheduled within $|\gamma_c| \leq 2 \cdot L_c^{FG}$ time slots, where γ_c is the **set of time slots** needed to schedule all edges in c .

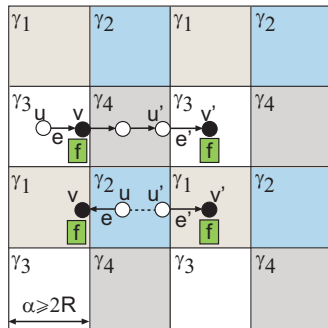
Phase 2: Time Slot Assignment

For the whole network:



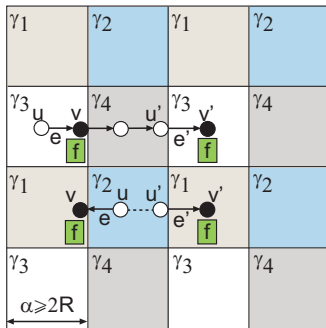
Phase 2: Time Slot Assignment

For the whole network:



Phase 2: Time Slot Assignment

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Lemma

The **minimum schedule length** for the whole network is bounded by $4 \cdot \max_c \{|\gamma_c|\}$, for all $\alpha \geq 2R$

Proof: Colors represent sets of **disjoint** time slots. Follows from **Vertex Color** on rectangular grid.

Summary on UDG

Algorithm $\mathcal{A}_{UDG}$

- 1 Run FREQUENCYGREEDY in each cell.
- 2 Run a greedy time slot assignment scheme.
e.g., schedule a maximal number of edges in each slot.

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K : number of frequencies

μ_α : maximum number of edges on the same frequency in any cell that can be scheduled simultaneously by any algorithm.

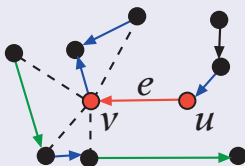
Approximation on Disk Graphs

Approximation on Disk Graphs

Notations

- $r(u)$: Transmission range of node u
- $\ell(e)$: Length of edge e
- $I(e)$: Set of edges that are either adjacent to e or form a secondary conflict with e
- $I_{\geq}(e)$: Subset of $I(e)$ that have larger disks than e

$$I_{\geq}(e) = \{e' = (u', v') : e' \in I(e), \ell(e') \geq \ell(e)\}$$



Approximation on Disk Graphs

Algorithm $\mathcal{A}_{DG}$ Overview

- **Phase 1:** Assign frequencies such that the maximum number of edges interfering with any given edge is minimized.
 - **0 – 1 Integer Linear Program:** Define indicator variables X_{vk} for edge $e = (u, v)$ as:

$$X_{vk} = \begin{cases} 1, & \text{if receiver } v \text{ is assigned frequency } f_k \\ 0, & \text{otherwise} \end{cases}$$

A frequency assignment is a 0 – 1 assignment to $X_{vk}, \forall e, \forall f_k$.

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- **Phase 2:** Sort edges in **non-increasing** order of **lengths**, and assign smallest available slots starting from the largest.

Approximation on Disk Graphs

0-1 Integer Linear Program of Frequency Assignment

Minimize λ

subject to :

$$\forall e = (u, v), \forall f_k : \sum_{v'} n(e, v') X_{vk} \leq \lambda \quad (1)$$

$$\forall e = (u, v) : \sum_{f_k} X_{vk} = 1, \quad (2)$$

$$\forall e = (u, v), \forall f_k : X_{vk} \in \{0, 1\} \quad (3)$$

- Solve the Linear Relaxation (LP) by modifying constraint (3)
- **Randomized Rounding:** Assign $Y_{vk} = 1$ with probability X_{vk}^* , where X_{vk}^* are the optimal fractional LP solutions, and Y_{vk} are the new integral random variables.

Approximation on Disk Graphs

Lemma

Let Y_{vk} be the rounded solution, as described above. Then,

$$\max_{e, f_k} \left\{ \sum_{e'=(u', v') \in I_{\geq}(e)} Y_{v'k} \right\} = O(\Delta(T) \log n \cdot \lambda^*),$$

with probability at least $(1 - 1/n)$.

Proof: Apply **Chernoff bound**.

Approximation on Disk Graphs

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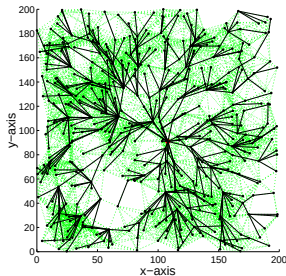
Theorem

The schedule constructed by time slot assignment strategy along with the frequency assignment using the above randomized rounding procedure results in a schedule of length $O(\Delta(T) \cdot \log n)$ times the optimum.

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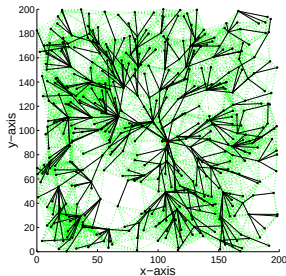
Properties of Spanning Trees



Shortest Path Tree (shallow and fat)

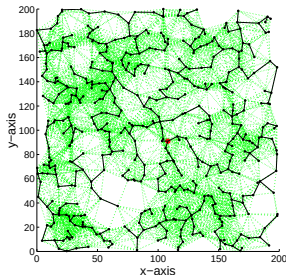
- High node degree $\Rightarrow$ low throughput
- Few hops $\Rightarrow$ low delay

Properties of Spanning Trees



Shortest Path Tree (shallow and fat)

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- Few hops $\Rightarrow$ low delay



Minimum Interference Tree (weighted MST) [Mobihoc '04] (deep and skinny)

$w(u, v)$: no. of nodes covered by the union of two disks centered at u and v , each of radius $|uv|$

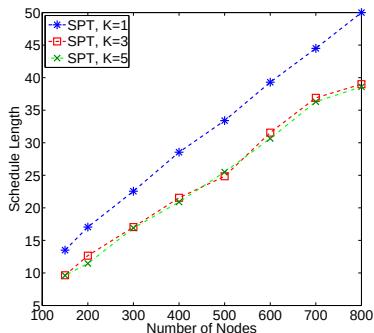
- Low node degree $\Rightarrow$ high throughput
- More hops $\Rightarrow$ high delay

Evaluation of $\mathcal{A}_{UDG}$ on SPT & MIT

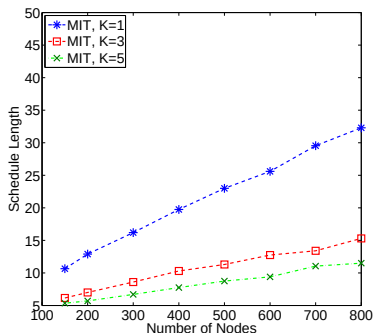
Parameters

Random geometric graph on a region of size 200×200 with $R = 25$

Shortest Path Tree



Minimum Interference Tree



Tree Property for Bounded Throughput

Theorem

If the **maximum node degree** of the routing tree is bounded by a **constant** $\Delta_C > 0$, then there exists an algorithm that gives a **constant factor** $8\mu_\alpha \cdot \Delta_C$ approximation on the optimal schedule length, where $\mu_\alpha > 0$ is a constant for a given cell size $\alpha \geq 2R$ and given node deployment.

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Properties

- Degree-bounded spanning trees? **Minimum Degree Spanning Tree** is NP-hard [SODA'92]
- Is **constant factor** approximation on **schedule length** and **delay** achievable?

Bicriteria Problems

Definition [Ravi et al. STOC '93]

Is a problem $\mathcal{P} = (\mathcal{X}, \mathcal{Y}, \mathcal{H})$, where $\mathcal{X}$ and $\mathcal{Y}$ are two objectives, and $\mathcal{H}$ is a subgraph.

$c_{\mathcal{X}}, c_{\mathcal{Y}}$: Cost functions on $\mathcal{X}$ and $\mathcal{Y}$

$B_{\mathcal{X}}$: Budget on $\mathcal{X}$ under $c_{\mathcal{X}}$

Goal: Find H minimizing $\mathcal{Y}$ under $c_{\mathcal{Y}}$ subject to $B_{\mathcal{X}}$

NP-Hard problems, especially when the objectives are conflicting.

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NP-Hard problems, especially when the objectives are **conflicting**.

Properties of Bicriteria Approximation [Marathe et al. JoA '08]

- **Robust:** Any (α, β) -approximation on $\mathcal{P} = (\mathcal{X}, \mathcal{Y}, \mathcal{H})$ can be transformed in **polynomial time** into a (β, α) -approximation on $\mathcal{P}' = (\mathcal{Y}, \mathcal{X}, \mathcal{H})$.
- **General:** Subsumes functional combinations of the two objectives.

Bicriteria Formulation

Bounded-Degree Minimum-Radius Spanning Tree

$\mathcal{P} = (\text{Degree, Radius, Spanning Tree})$

Radius: $R(T)$: Maximum hop distance from any node to sink in T

Δ^* : Budget on max degree

Goal: Find a spanning tree of min radius subject to max degree Δ^*

Bicriteria Formulation

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Goal: Find a spanning tree of **min radius** subject to **max degree** Δ^*

Approximation Definition

Algorithm $\mathcal{A}$ is an (α, β) -approximation to a bicriteria optimization problem $\mathcal{P} = (\text{Degree, Radius, Spanning Tree})$ if:

- $\Delta(T) \leq \alpha + \Delta^*$
- $R(T) \leq \beta \cdot R^*(T)$

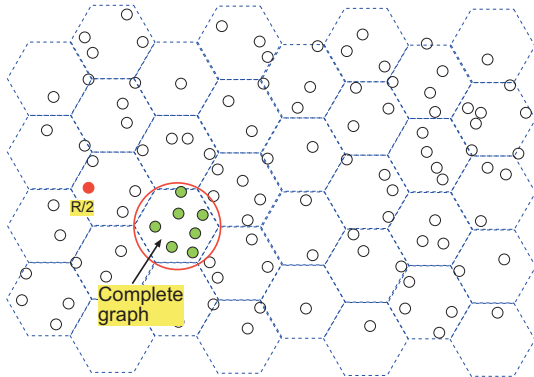
where $R^*(T)$ is the min radius of any spanning tree T whose max degree is Δ^* .

Algorithm Overview

Algorithm $\mathcal{A}_{BDMRST}$

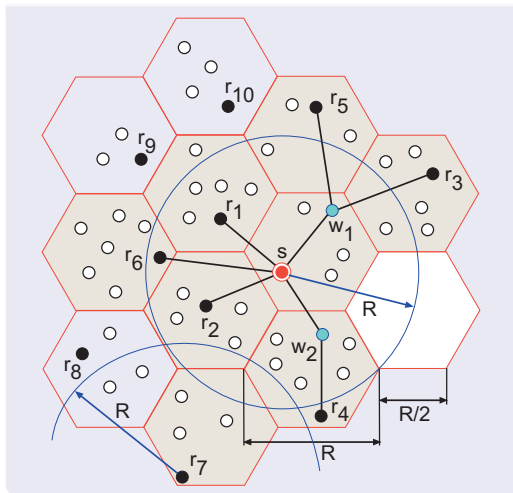
Three phases:

- 1 Construct a **Global Backbone Tree** of low radius.
- 2 Construct **degree-bounded Local Spanning Trees**.
- 3 **Merge** the local spanning trees and the global backbone tree.

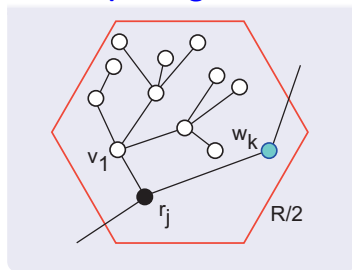


Algorithm Overview

Global Backbone Tree



Local Spanning Tree



Approximation Result & Evaluation

Theorem

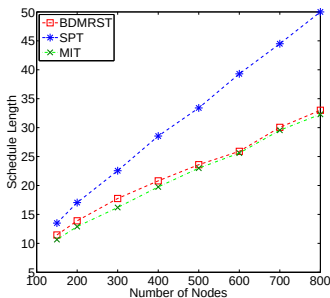
Algorithm $\mathcal{A}_{BDMRST}$ gives a **constant factor** (α, β) bicriteria approximation for the **Bounded-Degree Minimum-Radius Spanning Tree**, where $\alpha = 10$ and $\beta = 5$.

Approximation Result & Evaluation

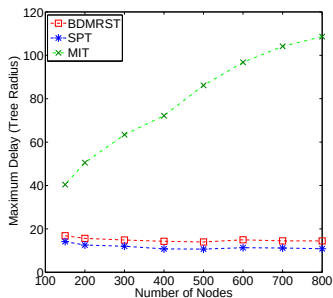
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Algorithm $\mathcal{A}_{BDMRST}$ gives a **constant factor** (α, β) bicriteria approximation for the **Bounded-Degree Minimum-Radius Spanning Tree**, where $\alpha = 10$ and $\beta = 5$.

Schedule Length



Max Delay



Outline

- 1 Maximizing Convergecast Throughput
 - Motivation and Preliminaries
 - Scheduling with Unlimited Frequencies
 - Approximations on Random Geometric Graphs
- 2 Throughput-Delay Trade-off
 - Motivation
 - Bicriteria Formulation
 - Optimal Spanning Trees
- 3 Industrial / Systems Research
 - Siemens Corporate Research
 - IBM Research
 - Other Experimental Research

My View of Systems Research

- **Ever-increasing gap** between theory and practice, especially in networking - need to bridge.
- **System building** should be motivated by solid **theoretical foundation**.
- **Theory** and **new insights** can also stem from systems research.
- **Practical, usable systems** that people could use - benefit to the society.
- A decade of sensor network research, but only a **handful of products**.
- **Satisfaction** and **fun** to see our own algorithms incorporated into commercial products.

RF Localization in Sensornets

Ecolocation (Error Controlling Localization) [Yedavalli IPSN '05]

- A **sequence-based**, **robust**, **RF** localization algorithm for sensornets
- **Relative** dependence on RSS (Received Signal Strength) readings, combating multi-path fading and shadowing
- Accuracy within a **few feet** in indoor environment with obstacles

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Experiments

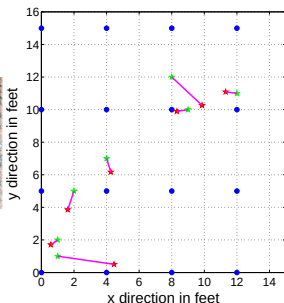
- Implementation in **NesC** on **Mica2** and **Micaz** motes running **TinyOS 1.x** and equipped with **802.15.4** radios.
- Implementation in **Java** on **Sun SPOT** motes.
- Several indoor experiments.
- Extension using **spatial diversity** techniques (e.g., Max Ratio Combiner).



RF Localization in Sensornets

Expt: Single Reference Node

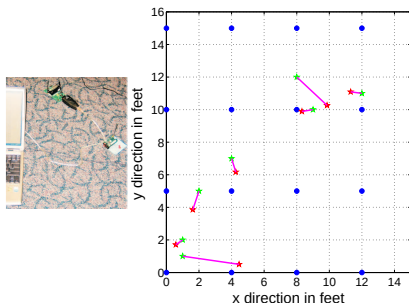
- 12×15 grid deployment of 16 nodes
- Localization space: RFID lab of size $19 \times 17 \text{ ft}^2$



RF Localization in Sensornets

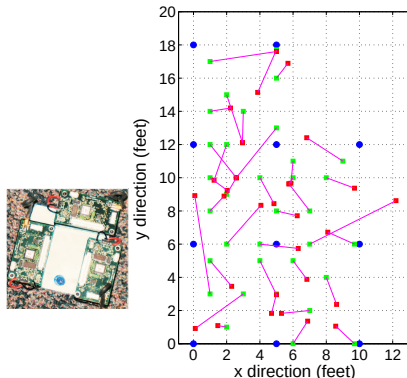
Expt: Single Reference Node

- 12×15 grid deployment of 16 nodes
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Expt: Clustered Reference Node

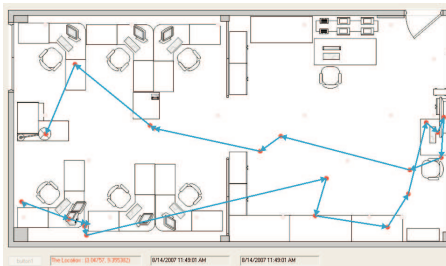
- 10×18 grid deployment of 11 clusters of 2 reference nodes



RF Localization in Sensor nets

Expt: Tracking a Moving Object

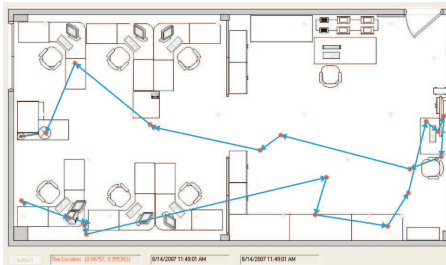
- Grid deployment of 20 reference nodes in both rooms of RFID lab
- Unknown node was carried in hand and moved in walking speed



RF Localization in Sensornets

Expt: Tracking a Moving Object

- Grid deployment of 20 reference nodes in both rooms of RFID lab
- Unknown node was carried in hand and moved in walking speed



Other Works

Proposed **security framework** for authentication, data confidentiality, and integrity of messages by applying **Elliptic Curve Cryptography (ECC)** and **μ -TESLA** broadcast authentication protocol.

QoE Characterization for VoD Services in WiMAX

Motivation

- Massive deployment of **triple** and **quadruple** services (broadband Internet, TV, telephone, wireless) over converged networks (4G)
- Guarantee subscribers' **Quality-of-Experience (QoE)** and provide differentiated services

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Problem Statement

Design and implement

- **Robust QoE models** mapping network impairments (e.g., bandwidth fluctuation, packet loss, delay, jitter) to user-perceived degradation
- **Orchestration mechanisms**, such as **bandwidth provisioning**, based on the above QoE model

to provide satisfactory user experience for **Video-on-Demand** services in **WiMAX**.

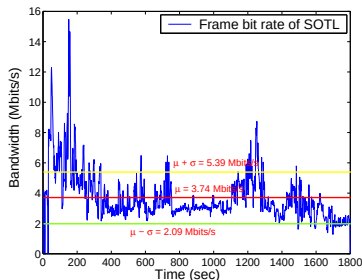
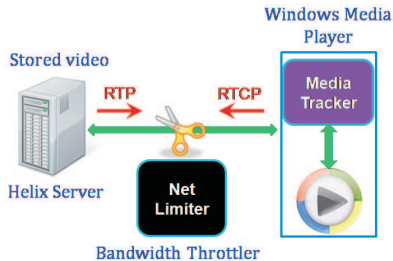


Figure: 30 mins of VBR traffic for Silence of the Lambs movie

QoE Characterization for VoD Services in WiMAX

Experiments to Gain Insights

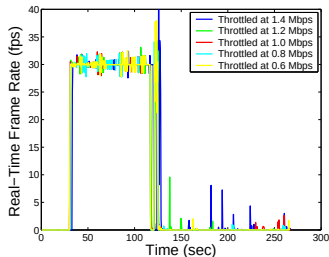
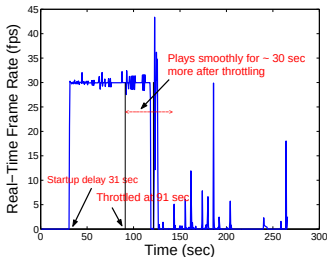
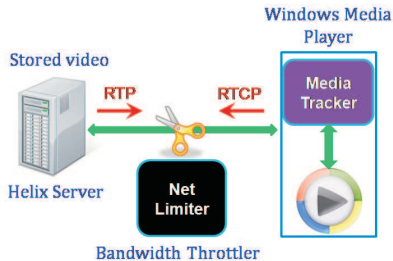
- Measure effects of initial client buffering and bandwidth throttling on streaming videos over LAN
- Smooth video playback for **initial buffering time** after stalling
- Small **BW variation** and **packet losses** cause major degradation in QoE



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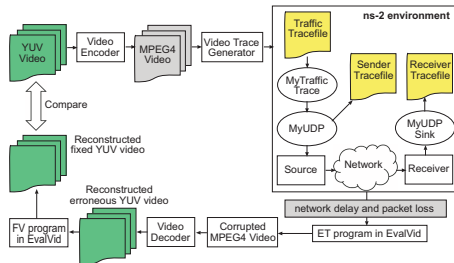
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QoE Characterization for VoD Services in WiMAX

Implementation on ns-2 + Evalvid

- Extract frame type, size, and inter-packet times from source video traces
- Fragment frames into smaller segments and transmit over wireless network
- Generate compressed MPEG4 video from received traces and original video
- Decode into YUV format and compute PSNR



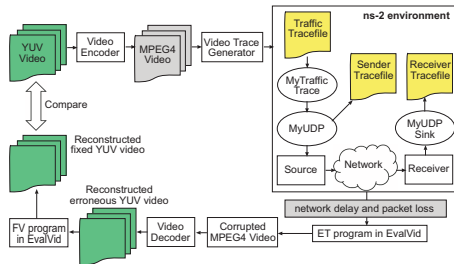
Sample Video Characteristics

MPEG4; VBR; CIF; GoP IPPBPPBPPBPPBPPB; $\mu_f = 0.22$ Mb;
 $\sigma_f = 0.12$ Mb; Total frames: 17681; i-frames:1106; p-frames:7735;
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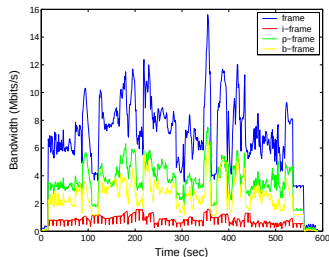
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QoE Characterization for VoD Services in WiMAX

Dynamic Bandwidth Provisioning

- Track cumulative moving average and standard deviation of source bit rate
- Provision BW at fixed intervals, such that BW provisioned at interval $(i + 1)$ can sustain a variation of k times the s.d. of BW of interval i

$$BW(i + 1) = \mu(i) + k \cdot \sigma(i) + \alpha/w - \beta \cdot \left[\sum_j w \cdot BW(j) - w \cdot i \cdot \mu(i) \right] / w$$

where, α : threshold buffer content
 $\mu(i), \sigma(i)$: cumulative moving average and s.d. until interval i
 w : interval duration
 β : constant parameter

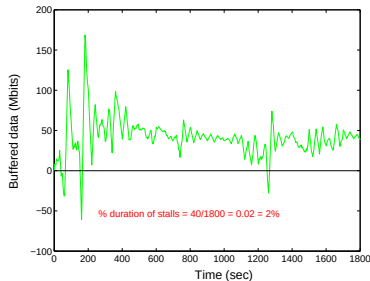
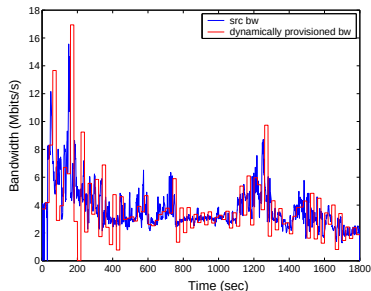
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Other Experimental / Industrial Research

Other Experimental / Industrial Research

Backpressure Collection Protocol (BCP) [Moeller IPSN'10]

- No explicit s-d path computation: Routing and forwarding decisions made independently for each packet by computing **queue differentials**
- **ns-2** implementation of dynamic backpressure routing protocol
- Understanding of **TinyOS 2.x** code and **Tutornet** testbed expts

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Ubiquitous Computing - Honeywell

- Design and implementation of **node deployment** and **location tracking** algorithms on **MSP430** microcontrollers
- Embedded programming and comparative study of **cryptographic ciphers** (AES, TEA)
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GSM Development - Lucent

- Design and implementation of **load distribution algorithms** for circuit switched and packet switched traffic in GSM base station
- Programming in C, C++, and Java for Lucent ATM and LAN switches

Other Experimental / Industrial Research

AT&T Labs Research (Summer '10)

- Extend analysis of **Limited Chain Relay (LCR)** algorithm for on-demand multimedia content distribution over communication networks
- Development of **Monte Carlo** simulation
- Analyze **network data** related to demand for various demand assets with particular attention to temporal and spatial contiguity in demands to assess savings in a realistic environment.

Conclusions

- **Constant factor** approx on UDG, and $\Delta(T) \cdot \log(n)$ approx on DG for maximizing convergecast throughput.
- **Bicriteria** formulation and constant factor approximation for throughput-delay trade-off.
- Implementation on **TinyOS 1.x** and **Sun SPOT** of a sequence-based, robust, RF localization algorithm.
- QoE modeling and **ns-2** implementation of bandwidth provisioning for VoD services in WiMAX networks.
- **ns-2** implementation of Backpressure Collection Protocol and study of **TinyOS 2.x** code.
- Embedded programming.

Acknowledgment

Collaborators

- Bhaskar Krishnamachari (USC)
- Ozlem Durmaz Incel (UTwente, Bogazici University, Turkey)
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- Shivkumar Kalyanaraman (IBM Research)
- Chellury Ram Sastry (Siemens Research)
- Yi Wang (USC)
- Krishnakant Chintalapudi (MSR)

Thank You!